INCH-POUND
MIL-DTL-12520/9B
31 March 2005
SUPERSEDING
MIL-C-12520/9A (EL)
5 November 1965

DETAIL SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL WATERPROOF (SHELL SIZE 20, FEMALE INSERTS)

Inactive for new design after 24 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-12520.

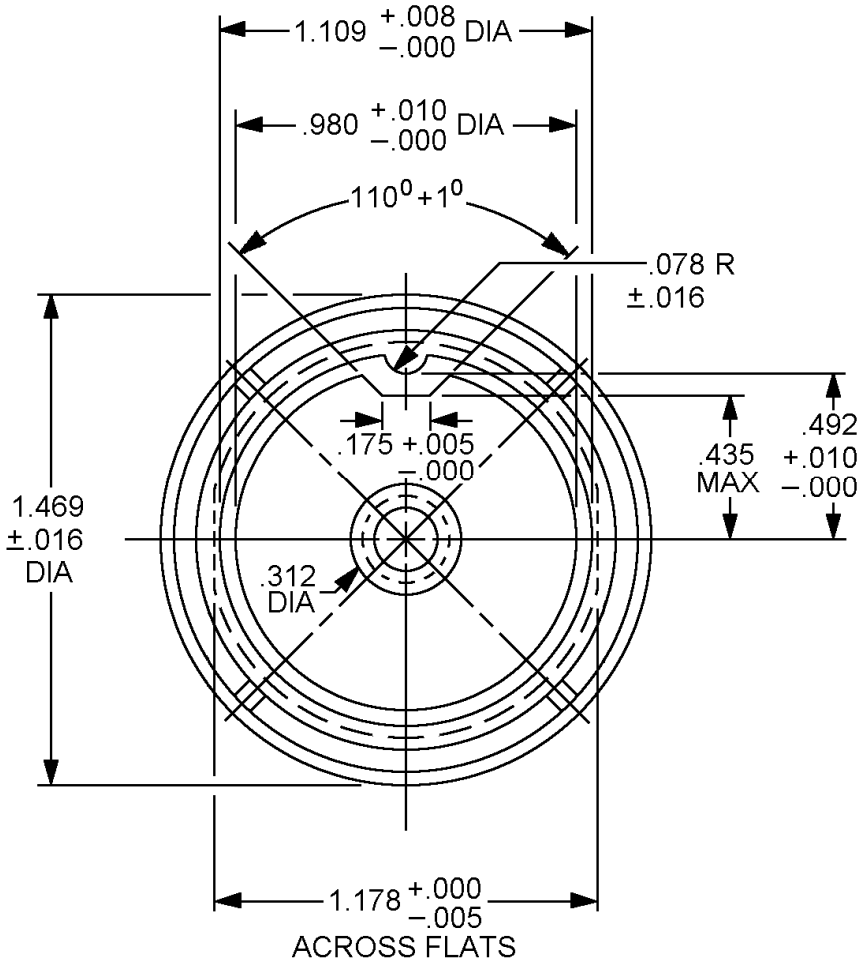
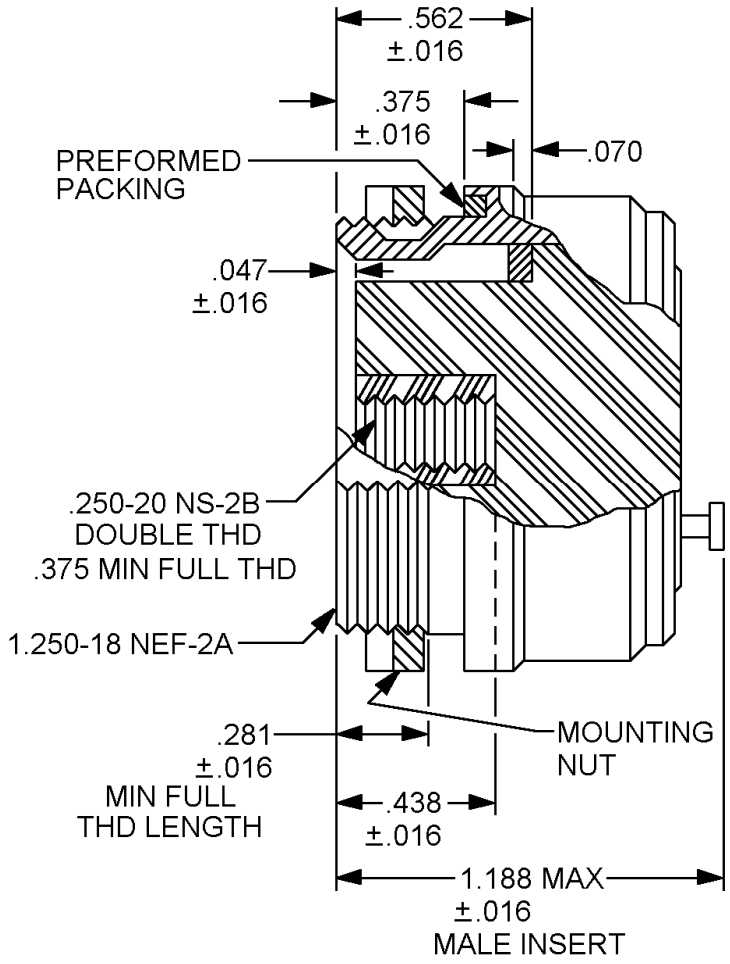


FIGURE 1. Receptacle shell and insert dimensions.

AMSC N/A FSC 5935

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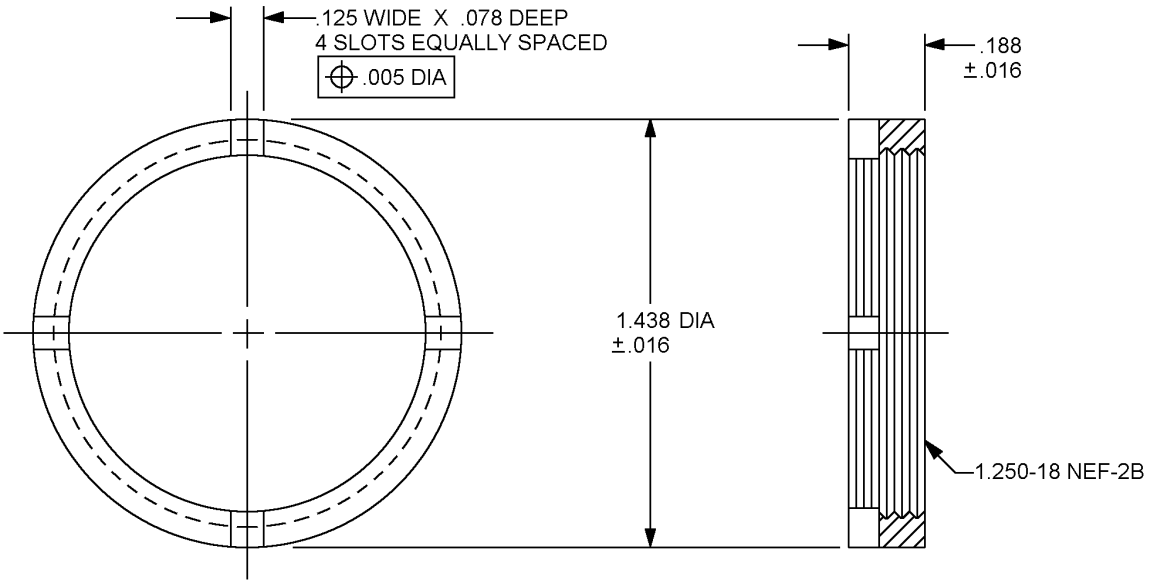


Inches	mm
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.047	1.19
.070	1.78
.078	1.98
.175	4.44
.281	7.14
.312	7.92
.375	9.52
.435	11.05
.438	11.12
.492	12.50
.562	14.27
.980	24.90
1.109	28.17
1.178	29.92
1.188	30.17
1.250	31.75
1.469	37.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm).

FIGURE 1. Receptacle shell and insert dimensions – Continued.

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Inches	mm
.005	0.13
.016	0.41
.078	1.98
.125	3.17
.188	4.77
1.250	31.75
1.438	36.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm).

FIGURE 2. Mounting nut dimensions.

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REQUIREMENTS

Connectors covered by this specification shall have the electrical ratings specified in table I.

TABLE I. Electrical ratings.

Connector type	Limiting operating voltage (sea level)		Nominal current
	AC	DC	Amperes
UW2020FA00	900	1,250	41
UW2020FB00	900	1,250	22

Dimensions and configurations: See figures 1 and 2.

Design and construction: Connectors shall be of the design, construction, and physical dimensions specified on figure 1.

Mounting nut: The mounting nut shall be as specified on figure 2. Material shall be brass conforming to ASTM-B36 and ASTM-B121, and shall be cadmium plated in accordance with class 1, type II, of SAE-AMS-QQ-P-416. The resulting finish shall be olive drab in color.

This receptacle connector mates with MIL-C-12520/6 and MIL-C-12520/17.

Dielectric withstanding voltage: Test in accordance with MIL-DTL-12520, except that the test potential shall be 2,800 volts (rms).

Voltage drop: The voltage drop shall not exceed 20 millivolts for insert-arrangement A or 21 millivolts for insert-arrangement B. The test current for connectors shall be 35 amperes DC for insert-arragement A, and 20 amperes DC for insert-arrangement B.

Cover and chain assemblies: Cover and chain assemblies are in accordance with style 2X of MIL-C-12520/12.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Reference documents. In addition to MIL-DTL-12520, this document references the following:

ASTM-B36
ASTM-B121
MIL-C-12520/6
MIL-C-12520/12
MIL-C-12520/17
SAE-AMS-QQ-P-416

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Navy - AS
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4726-002)

Review activities:
Navy - MC
Air Force - 99

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